UNIT - III

Numerical Solution of Ordinary Differential Equations

(Initial Value Problems: Taylor, Picard, Euler, Modified Euler, Runge-Kutta)

1. Introduction

Many engineering problems lead to a first-order differential equation with an initial condition, called an INITIAL VALUE PROBLEM (IVP):

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0$$

- *Analytical solutions exist only for special forms. When no closed form is available we compute y at a set of points $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, ...NUMERICALLY (h = step size).*
- *SINGLE-STEP methods (this unit) compute y_{n+1} from y_n only: Taylor, Picard, Euler, Modified Euler, Runge-Kutta.*
- *Accuracy improves when h is smaller, or when the method is of higher order. A method has order p if its local error is $O(h^{p+1})$.*

Notation: $x_n = x_0 + nh$, $y_n \approx y(x_n)$, $f_n = f(x_n, y_n)$.

2. Taylor's Series Method

If $y(x)$ is analytic at x_0, expand it about x_0:

Taylor's series solution

$$y(x) = y_0 + (x - x_0)\,y_0' + \frac{(x-x_0)^2}{2!}\,y_0'' + \frac{(x-x_0)^3}{3!}\,y_0''' + \cdots$$

- *Obtain* $y_0' = f(x_0, y_0)$ *from the equation, and* y'', y''', … *by DIFFERENTIATING the equation repeatedly, then put* $x = x_0,\ y = y_0$.
- *To go further, treat* (x_1, y_1) *as a new starting point and expand again:* $y_{n+1} = y_n + hy_n' + \frac{h^2}{2!}y_n'' + \cdots$
- *Merits: accurate if many terms are taken. Demerit: derivatives become tedious.*

Example 1. $\frac{dy}{dx} = x^2 + y^2,\ y(0) = 1$ **: find** $y(0.1)$ **and** $y(0.2)$ **(up to** x^4 **term)**

$y' = x^2 + y^2 \ \Rightarrow y'(0) = 0 + 1 = 1$

$y'' = 2x + 2yy' \ \Rightarrow y''(0) = 0 + 2(1)(1) = 2$

$y''' = 2 + 2(y'^2 + yy'') \ \Rightarrow y'''(0) = 2 + 2(1+2) = 8$

$y^{iv} = 2(2y'y'' + y'y'' + yy''') = 6y'y'' + 2yy''' \ \Rightarrow$

$y^{iv}(0) = 12 + 16 = 28$

$$y(x) = 1 + x + \frac{2x^2}{2!} + \frac{8x^3}{3!} + \frac{28x^4}{4!} = 1 + x + x^2 + \frac{4}{3}x^3 + \frac{7}{6}x^4 + \cdots$$

$y(0.1) = 1 + 0.1 + 0.01 + 0.001333 + 0.000117 = 1.1115;$

$y(0.2) = 1 + 0.2 + 0.04 + 0.010667 + 0.001867 = 1.2525.$

Example 2. $y' = x + y,\ y(0) = 1$ **: find** $y(0.1)$ **by Taylor's method**

$y' = x + y \Rightarrow 1; \quad y'' = 1 + y' \Rightarrow 2; \quad y''' = y'' \Rightarrow 2; \quad y^{iv} = 2$ *(all at* $x = 0$*).*

$$y(x) = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \cdots$$

$y(0.1) = 1 + 0.1 + 0.01 + 0.000333 + 0.0000083 = 1.110342$ *(exact* $2e^{0.1} - 1.1 = 1.110342$*).*

3. Picard's Method of Successive Approximations

Integrate $\dfrac{dy}{dx} = f(x, y)$ *from* x_0 *to* x *:* $y = y_0 + \displaystyle\int_{x_0}^{x} f(x, y)\, dx$*. The unknown* y *on the right is replaced step by step:*

Picard's iteration

$$y^{(1)} = y_0 + \int_{x_0}^{x} f(x, y_0)\, dx, \qquad y^{(2)} = y_0 + \int_{x_0}^{x} f(x, y^{(1)})\, dx$$

$$y^{(n+1)} = y_0 + \int_{x_0}^{x} f(x, y^{(n)})\, dx, \qquad n = 0, 1, 2, \ \ldots$$

- *Start with* $y^{(0)} = y_0$ *(a constant). Stop when two successive approximations agree to the required accuracy in the interval.*
- *Merit: gives* y *as a function of* x*. Demerit: integrals become difficult for complicated* f*.*

Example 3. $y' = x + y,\ y(0) = 1$ **: find** $y^{(1)}, y^{(2)}, y^{(3)}$ **and** $y(0.1)$

$$y^{(1)} = 1 + \int_0^x (x + 1)\, dx = 1 + x + \frac{x^2}{2}$$

$$y^{(2)} = 1 + \int_0^x \left(x + 1 + x + \frac{x^2}{2}\right) dx = 1 + x + x^2 + \frac{x^3}{6}$$

$$y^{(3)} = 1 + \int_0^x \left(1 + 2x + x^2 + \frac{x^3}{6}\right) dx = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}$$

$y(0.1) \approx 1.110337$ *(exact 1.110342).*

Example 4. $y' = x^2 + y^2,\ y(0) = 0$ **: find** $y(0.4)$ **using Picard**

$$y^{(1)} = \int_0^x x^2 dx = \frac{x^3}{3}\ ;\ y^{(2)} = \int_0^x \left(x^2 + \frac{x^6}{9}\right) dx = \frac{x^3}{3} + \frac{x^7}{63}$$

$$y^{(3)} = \frac{x^3}{3} + \frac{x^7}{63} + \frac{2x^{11}}{2079} + \frac{x^{15}}{59535}$$

$y(0.4) \approx 0.02136$ *(two terms:* $0.02133 + 0.00026$*). Four-decimal accuracy: 0.0213.*

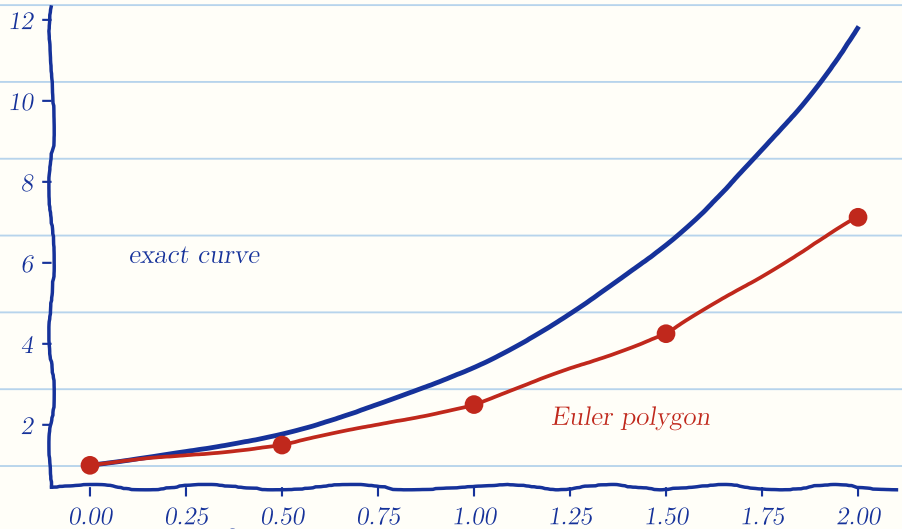
4. Euler's Method

Simplest method. On $[x_n, x_{n+1}]$ *replace the solution curve by its TANGENT at* (x_n, y_n) *(slope* $f(x_n, y_n)$*):*

Euler's formula

$$y_{n+1} = y_n + h\, f(x_n, y_n), \qquad n = 0, 1, 2, \ \ldots$$

Euler: tangent steps lag below the curve



- *Local error* $O(h^2)$*, global error* $O(h)$ *- FIRST order. Needs small* h*.*
- *Geometrically the curve is replaced by a polygon; error accumulates and for convex solution curves Euler values fall below the true values.*

Example 5. $\frac{dy}{dx} = x + y,\ y(0) = 1$ **: find** $y(0.5)$ **taking** $h = 0.1$

n	x_n	y_n	$f = x_n + y_n$	$y_{n+1} = y_n + 0.1f$
0	0.0	1.0000	1.0000	1.1000
1	0.1	1.1000	1.2000	1.2200
2	0.2	1.2200	1.4200	1.3620
3	0.3	1.3620	1.6620	1.5282
4	0.4	1.5282	1.9282	1.7210

$y(0.5) \approx 1.7210$ *by Euler; exact* $= 2e^{0.5} - 1.5 = 1.7974$. *Error* ≈ 0.0764 *(large - so refine).*

Example 5(b). $\frac{dy}{dx} = \frac{y - x}{y + x},\ y(0) = 1$ **: find** $y(0.1)$ **taking** $h = 0.02$

n	x_n	y_n	$f(x_n, y_n)$	y_{n+1}
0	0.00	1.00000	1.00000	1.02000
1	0.02	1.02000	0.96154	1.03923
2	0.04	1.03923	0.92587	1.05775
3	0.06	1.05775	0.89264	1.07560
4	0.08	1.07560	0.86154	1.09283

$y(0.1) \approx 1.0928$.

Where does Euler's formula come from?

Taylor: $y(x_n + h) = y_n + hy'_n + \frac{h^2}{2}y''_n + \cdots$. *Keeping only the first two terms gives Euler; the dropped term* $\frac{h^2}{2}y''(\xi)$ *is the LOCAL TRUNCATION ERROR. Halving* h *halves the final error but doubles the number of steps.*

5. Modified Euler's Method (Heun / Predictor-Corrector)

Euler uses the slope at the LEFT end only. Improve by using the AVERAGE of slopes at both ends (trapezoidal rule):

Modified Euler formulae

$$\text{Predictor:} \quad y_{n+1}^{(0)} = y_n + h\, f(x_n, y_n)$$

$$\text{Corrector:} \quad y_{n+1}^{(k+1)} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f\left(x_{n+1}, y_{n+1}^{(k)}\right)\right]$$

- *Repeat the corrector for $k = 0, 1, 2, \ldots$ until two successive values of y_{n+1} agree.*
- *Second-order method: local error $O(h^3)$, global error $O(h^2)$.*

Example 6. $y' = x + y,\ y(0) = 1$ **: find** $y(0.1)$ **and** $y(0.2)$ $(h = 0.1)$

Value	Computation	Result
$y_1^{(0)}$	*1.00000 + 0.1(1.00000)*	*1.10000*
$y_1^{(1)}$	*avg slope with 1.10000*	*1.11000*
$y_1^{(2)}$	*avg slope with 1.11000*	*1.11050*
$y_2^{(0)}$	*1.11050 + 0.1(1.21050)*	*1.23155*
$y_2^{(1)}$	*avg slope with 1.23155*	*1.24260*
$y_2^{(2)}$	*avg slope with 1.24260*	*1.24316*

$y(0.1) \approx 1.11000$, $y(0.2) \approx 1.24316$ *(exact 1.24281). Much better than Euler.*

Example 7. Compare errors at $x = 0.5$: $y' = x + y$, $y(0) = 1$, $h = 0.1$

Method	Order	$y(0.5)$	Error
Euler	*1*	*1.721020*	*0.076423*
Modified Euler (1 corr.)	*2*	*1.794894*	*0.002549*
RK 4th order	*4*	*1.797441*	*1.26e-06*
Exact	-	*1.797443*	*0*

6. Runge-Kutta Methods

Taylor-quality accuracy WITHOUT computing higher derivatives: use several slope evaluations $k_1, k_2, \ldots$ in each step.

(a) Second-order Runge-Kutta

RK-2 (improved Euler)

$$k_1 = h\, f(x_n, y_n), \qquad k_2 = h\, f(x_n + h,\ y_n + k_1)$$

$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$$

(Another RK-2 form uses the mid-point: $k_2 = hf(x_n + \frac{h}{2}, y_n + \frac{k_1}{2})$, $y_{n+1} = y_n + k_2$.) Local error $O(h^3)$.

(b) Fourth-order Runge-Kutta (the most widely used)

RK-4 formulae

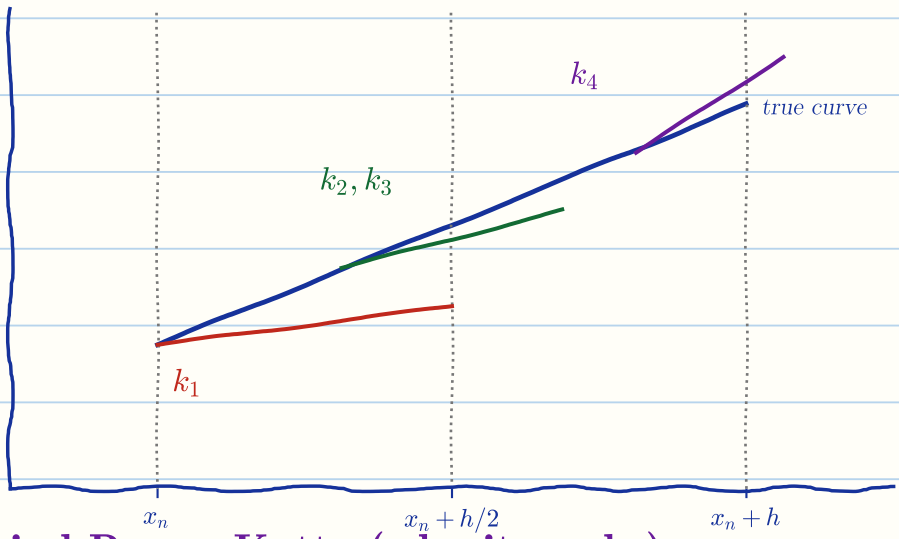
$$k_1 = hf(x_n, y_n), \qquad k_2 = hf\left(x_n + \frac{h}{2},\ y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{h}{2},\ y_n + \frac{k_2}{2}\right), \qquad k_4 = hf(x_n + h,\ y_n + k_3)$$

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

- *Local error $O(h^5)$, global error $O(h^4)$. Needs 4 function values per step but NO derivatives of f.*
- *k_1: slope at start, k_2, k_3: slopes at the mid-point (two estimates), k_4: slope at the end. Weights 1:2:2:1.*

RK-4: four slopes are sampled and averaged



Idea behind Runge-Kutta (why it works)

Assume $y_{n+1} = y_n + ak_1 + bk_2$ with $k_1 = hf(x_n, y_n)$, $k_2 = hf(x_n + \alpha h,\ y_n + \beta k_1)$. Expand k_2 by Taylor and match with the Taylor series of y up to h^2 terms: this forces

$$a + b = 1, \qquad b\alpha = \frac{1}{2}, \qquad b\beta = \frac{1}{2}$$

Choosing $a = b = \frac{1}{2}$, $\alpha = \beta = 1$ gives RK-2 above; choosing

$a=0, b=1, \alpha=\beta=\frac{1}{2}$ *gives the mid-point form. RK-4 matches Taylor up to* h^4 *in the same way.*

Example 8. RK-2: $y'=x+y,\ y(0)=1$ **: find** $y(0.2)$ **taking** $h=0.2$

$k_1=0.2(0+1)=0.2000$; $k_2=0.2\,f(0.2,\ 1.2)=0.2(1.4)=0.2800$

$y(0.2)=1+\frac{1}{2}(0.20+0.28)=1.2400$ *(exact 1.2428).*

Example 9. RK-4: $y'=x+y,\ y(0)=1$ **: find** $y(0.2)$ **taking** $h=0.2$

$k_1=0.2(0+1)=0.2000$

$k_2=0.2\,f(0.1,\ 1+0.10)=0.2(0.1+1.1)=0.2400$

$k_3=0.2\,f(0.1,\ 1+0.12)=0.2(0.1+1.12)=0.2440$

$k_4=0.2\,f(0.2,\ 1+0.244)=0.2(0.2+1.244)=0.2888$

$y(0.2)=1+\frac{1}{6}(0.20+2(0.24)+2(0.244)+0.2888)=1.24280$ *(exact 1.24281).*

Example 10. RK-4: $y'=\dfrac{y^2-x^2}{y^2+x^2},\ y(0)=1$ **: find** $y(0.2)$ **with** $h=0.2$

	Expression	Value
k_1	$h\,f(0,1)$	*0.20000*
k_2	$h\,f(0.1,\ 1+k_1/2)$	*0.19672*
k_3	$h\,f(0.1,\ 1+k_2/2)$	*0.19671*
k_4	$h\,f(0.2,\ 1+k_3)$	*0.18913*

$y(0.2)=1+\frac{1}{6}(k_1+2k_2+2k_3+k_4)=1.19600.$

(High-accuracy check: 1.19601.)

Example 11. RK-4 for two steps: $y' = x + y,\ y(0) = 1,\ h = 0.1$: find $y(0.2)$

x_n	y_n	k_1	k_2	k_3	k_4	y_{n+1}
0.0	1.00000	0.10000	0.11000	0.11050	0.12105	1.11034
0.1	1.11034	0.12103	0.13209	0.13264	0.14430	1.24281

$y(0.2) = 1.24281$ *agrees with exact 1.24281 to 5 decimals.*

Example 12. RK-4: $y' = 3x + \dfrac{y}{2},\ y(0) = 1$: find $y(0.2)$ with $h = 0.2$

k_1	k_2	k_3	k_4	$y(0.2)$
0.10000	0.16500	0.16825	0.23683	1.16722

Here $k_2 = 0.2\, f(0.1,\ 1.05) = 0.2(0.3 + 0.525),\ k_3 = 0.2\, f(0.1,\ 1 + k_2/2),$ $k_4 = 0.2\, f(0.2,\ 1 + k_3).$

Choosing the step size / checking accuracy

- *Compute $y(x)$ with step h and again with $h/2$. If the two answers agree to the required decimals, accept; otherwise halve again.*
- *For RK-4 the error of the $h/2$ result is about $\frac{1}{16}$ of the error of the h result (order 4); for Euler only $\frac{1}{2}$.*
- *If f changes rapidly (stiff problems) a very small h is needed or implicit methods are used (beyond this syllabus).*

7. Comparison of Methods

Method	Order	Evaluations / step	Remarks
Taylor	depends on terms	1 (+ derivatives)	needs $y', y'', \ldots$
Picard	series in x	integrals	gives $y(x)$
Euler	1	1	simple, least accurate
Modified Euler	2	2 (per iteration)	predictor-corrector
RK-2	2	2	no derivatives
RK-4	4	4	most popular, accurate

Formula sheet

Taylor: $y = y_0 + hy_0' + \frac{h^2}{2!}y_0'' + \cdots$ Picard:

$y^{(n+1)} = y_0 + \int f(x, y^{(n)})dx$

Euler: $y_{n+1} = y_n + hf_n$; Modified Euler:

$y_{n+1} = y_n + \frac{h}{2}[f_n + f(x_{n+1}, y_{n+1}^{(0)})]$

RK-4: $y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

- In exams show a neat table of $k_1 \ldots k_4$; keep at least 4-5 decimal places; always state the final y with units of accuracy asked.
- Typical questions: Taylor/Picard (5 marks), Euler or Modified Euler (5 marks), RK-4 (10 marks).

Practice problems

1. Use Euler's method: $y' = 1 + y^2$, $y(0) = 0$, find $y(0.6)$ with $h = 0.2$. Ans: ≈ 0.6413
2. Picard: $y' = x - y^2$, $y(0) = 1$, find $y^{(2)}(x)$ and $y(0.1)$. Ans: $1 - x + \frac{3x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} - \frac{x^5}{20}$; $y(0.1) \approx 0.9144$
3. Modified Euler: $y' = \log_{10}(x + y)$, $y(0) = 2$, $h = 0.2$, find $y(0.2)$ [Ans: 2.0656].

4. *RK-4:* $y' = x^2 + y^2,\ y(1) = 1.5,\ h = 0.1$*: find* $y(1.1)$ *[Ans: 1.8955].*